Amortized Efficiency for Pairing Delegation

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- **New hardness assumption (DBDH)**: Given $(g_1, g_1^a, g_2, g_2^b, g_1^c, T)$ with $g_i \in \mathbb{G}_i$, hard to decide whether $T = e(g_1, g_2)^{abc}$ or random.

Pairing-based Cryptography

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Joux's tripartite Diffie-Hellman: Alice broadcasts $[a]\,P$, Bob broadcasts $[b]\,P$, and Charlie broadcasts $[c]\,P$. Each then computes the shared key

$$K = e(P,P)^{abc} = \underset{\substack{\uparrow \\ \text{Alice}}}{e([b]P,[c]P)^{a}} = \underset{\substack{\uparrow \\ \text{Bob}}}{e([a]P,[c]P)^{b}} = \underset{\substack{\uparrow \\ \text{Charlie}}}{e([a]P,[b]P)^{c}}$$

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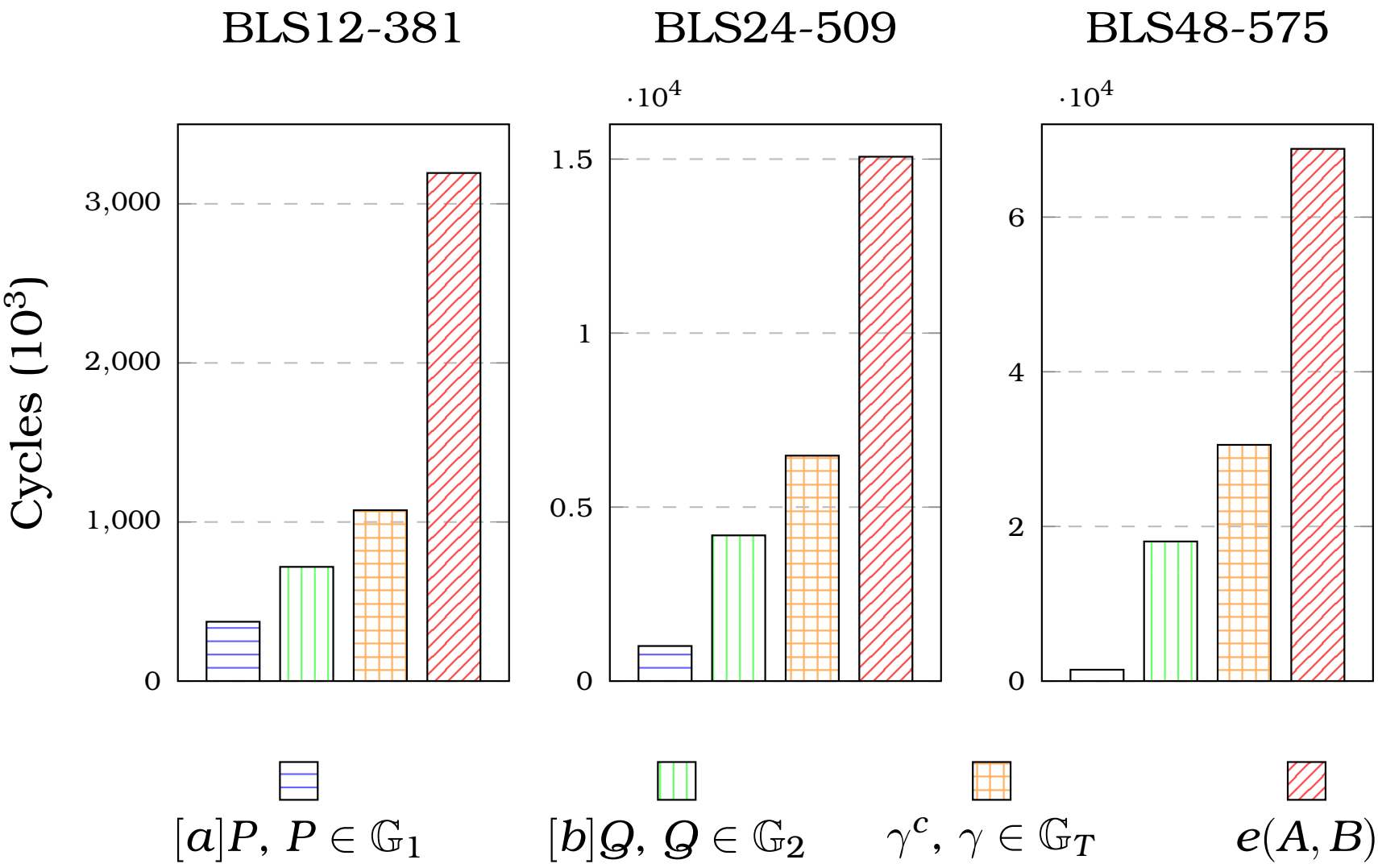
- **zkSNARKs (Groth16)**: Constant-size zkSNARK proofs (3 group elements) with verification reducible to 3 pairings.

Computational Costs

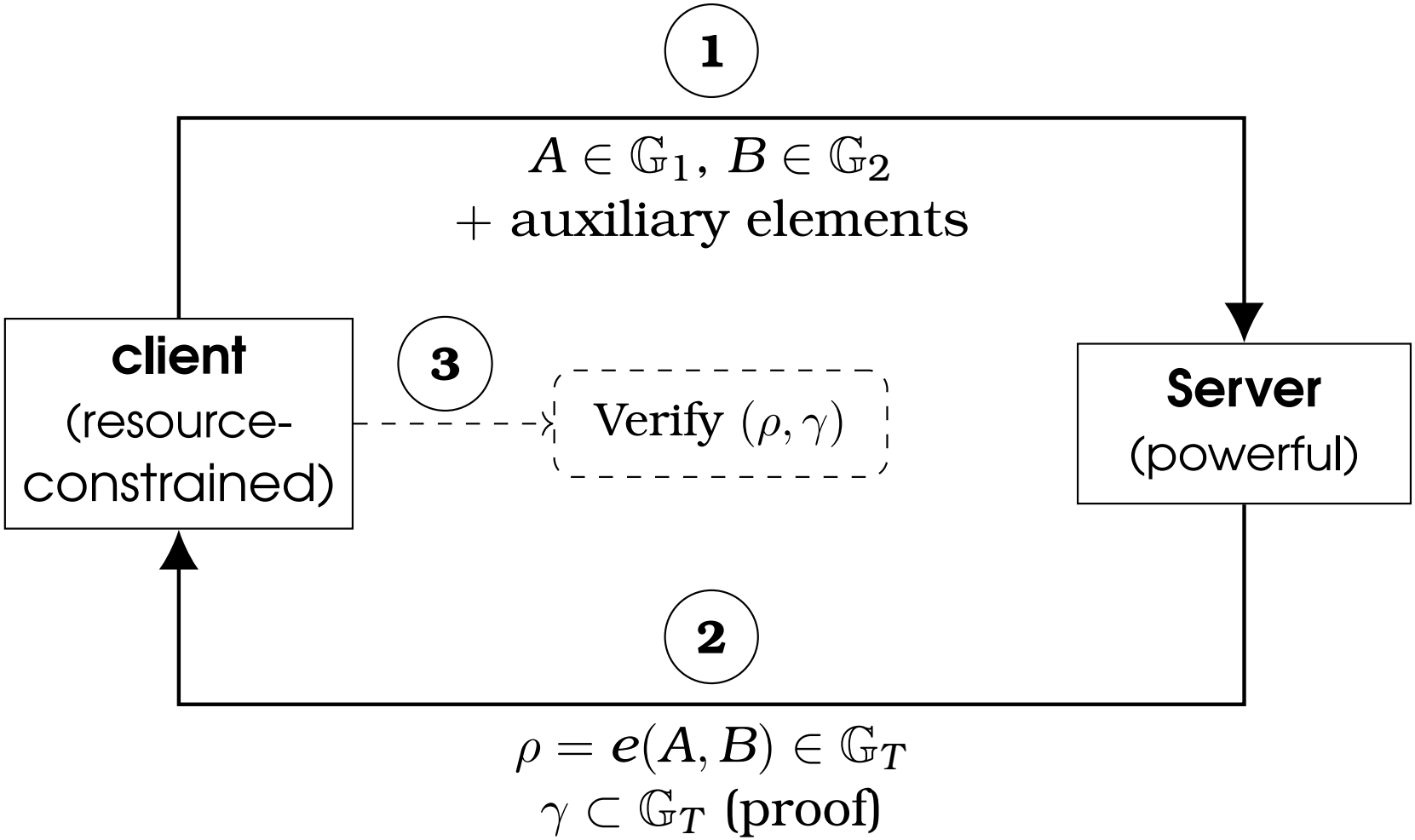


Pairing Delegation

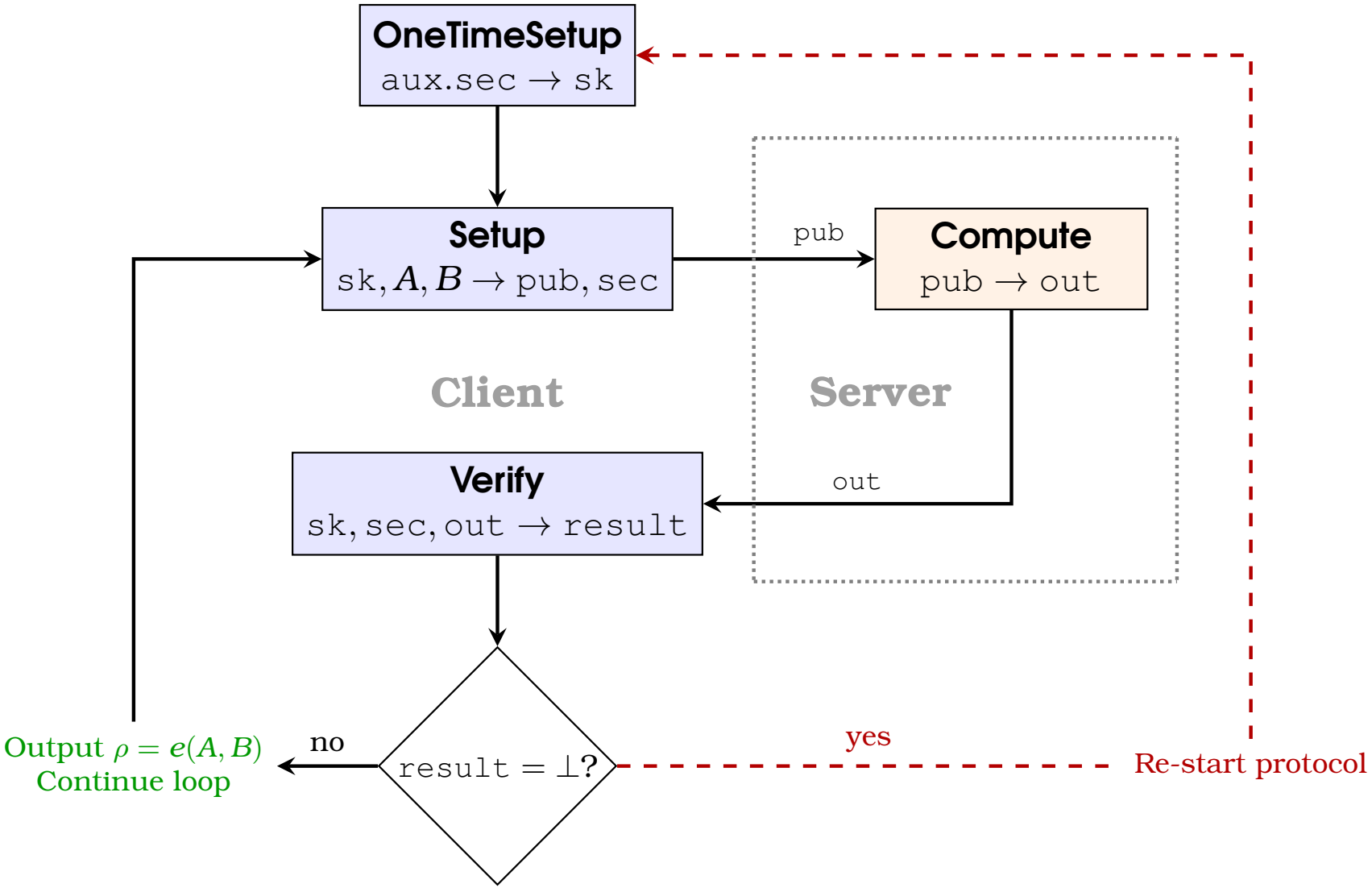
All pairing delegation protocols are

- One-shot:
- Proven unconditionally secure $\rightarrow$ This

One-Shot Pairing Delegation



Sequential Pairing Delegation



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- **Correctness**: If the server behaves honestly, the output of the verification phase is the intended (vector of) pairing(s).

$$\Pr\left[\ \mathtt{result} = e(A,B)\ \middle|\ \begin{array}{c}(\mathtt{pub},\mathtt{sec}) \leftarrow \mathsf{Setup}(\mathtt{sk},A,B)\\ \mathtt{out} \leftarrow \mathsf{Compute}(\mathtt{pub})\\ \mathtt{result} \leftarrow \mathsf{Verify}(\mathtt{sk},\mathtt{sec},\mathtt{out})\end{array}\right] = 1\,.$$

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- **Amortized Efficiency**: The client's workload when delegating L pairings must be lower than computing them locally.

$$\frac{\mathsf{cost}(\mathsf{OneTimeSetup}) + L \cdot (\mathsf{cost}(\mathsf{Setup}) + \mathsf{cost}(\mathsf{Verify}))}{L} \leq \mathsf{cost}(e(A,B))$$

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This guarantees unconditional security. Why?

One way to design this

Lemma 1 (Identifying forgeries)

Let $\mathcal{A}$ be an algebraic adversary, $(\xi, r) \in \mathbb{G}_T \times \mathbb{Z}_q^*$ the client's secret values, $\mathtt{out} = (\rho, \gamma) \in \mathbb{G}_T \times \mathbb{G}_T$ the output of an honest server, and Verify *a verification phase consisting of the checks*

$$\left(\xi \stackrel{?}{=} \rho^r \cdot \gamma\right) \wedge \left(\rho \stackrel{?}{\in} \mathbb{G}_T\right) . \tag{1}$$

If $(\mathcal{A}(\mathtt{pub}) \to \mathtt{out}^* \neq \mathtt{out}) \wedge (\mathsf{Verify}(\mathtt{out}^*) \neq \perp)$ then $\mathcal{A}$ outputs the secret r.

Proof.

By the closure property of $\mathbb{G}_T$, if $\left(\xi = (\rho^*)^r \cdot (\gamma^*)\right) \wedge (\rho^* \in \mathbb{G}_T)$, then $\mathtt{out}^*$ can be written as a deviation from the honest output: $\rho^* = \rho \cdot \eta$ and $\gamma^* = \gamma \cdot \eta^{-r}$ for any public element $\eta \in \mathbb{G}_T$. Indeed, $(\rho^*)^r \cdot (\gamma^*) = (\rho \cdot \eta)^r \cdot \gamma \cdot \eta^{-r} = \rho^r \cdot \gamma = \xi$. □

One way to design this

For **amortized efficiency**:

- Sequential delegation: Reuse ξ by generating in the next delegations secrets (r_i, u_i, v_i) such that $u_i \cdot v_i = u_j \cdot v_j$. The public values will still be equipped with an extra degree of freedom.
- Sample short scalars. Short $\mathbb{G}_T$ exponentation is (a lot) cheaper than a full-range one.

Wait a minute... do we still get unconditional security?

Meet-in-the-middle attack

A toy example:

Let $(\mathbb{G} = \langle P \rangle, +)$ be a cyclic group of prime order q, and (r_1, r_2, u) a tuple of secret values such that $r_1, r_2 \xleftarrow{\$} S \subset \mathbb{Z}_q^*$ and $u \xleftarrow{\$} \mathbb{Z}_q^*$. Let $U = [u]P \in \mathbb{G}$, and consider the public view

$$\mathtt{pub} = \begin{cases} C = [r_1]U + X \\ D = [r_2]U + Y \end{cases}, \tag{2}$$

where $X, Y \in \mathbb{G}$ are public and $X \neq C, Y \neq D$. $\mathcal{A}$ can attempt to learn the secret tuple from (2) by picking an element

$$U^* \in \mathcal{I} = \left\{[t_1^{-1}](C - X) \colon t_1 \in S\right\} \cap \left\{[t_2^{-1}](D - Y) \colon t_2 \in S\right\} \tag{3}$$

and its corresponding indexes r_1^*, r_2^*, which takes up to $2 \cdot |S|$ computations.

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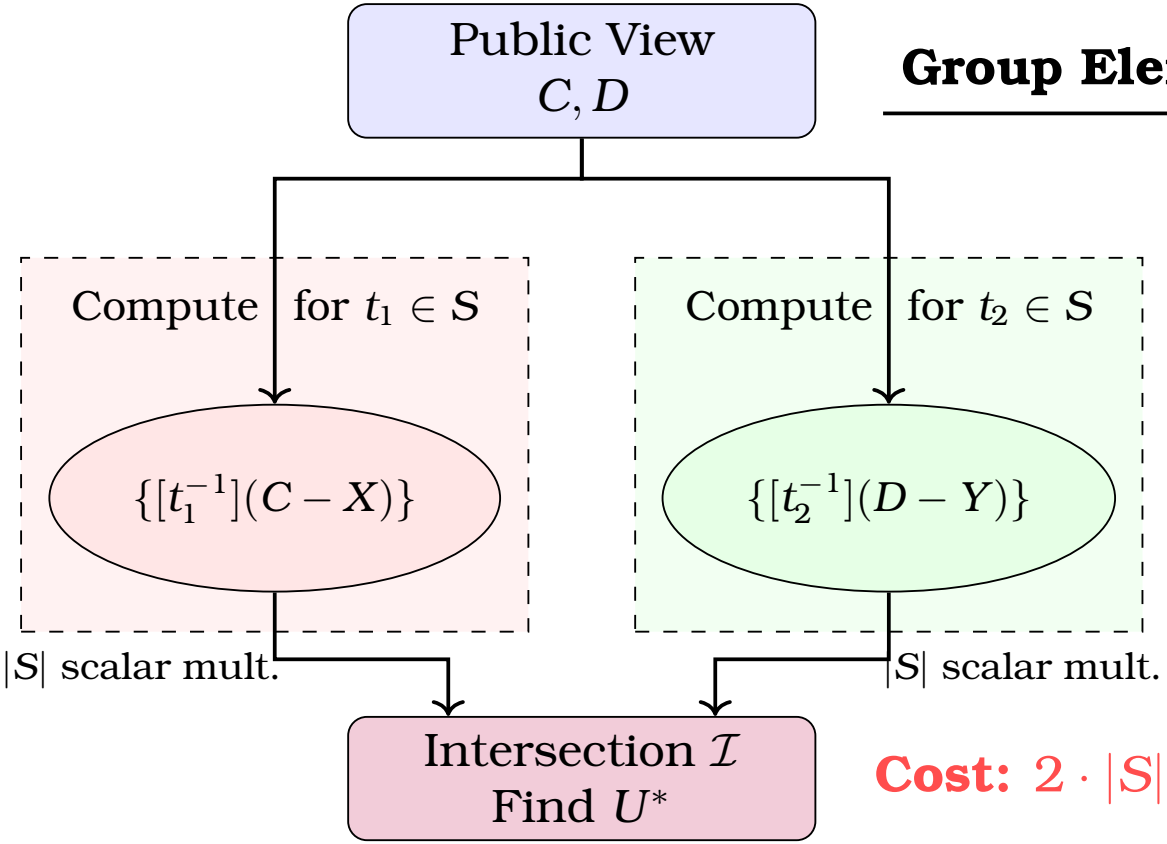

Group Element Intersection Attack

Cost: $2 \cdot |S|$ **scalar multiplications**

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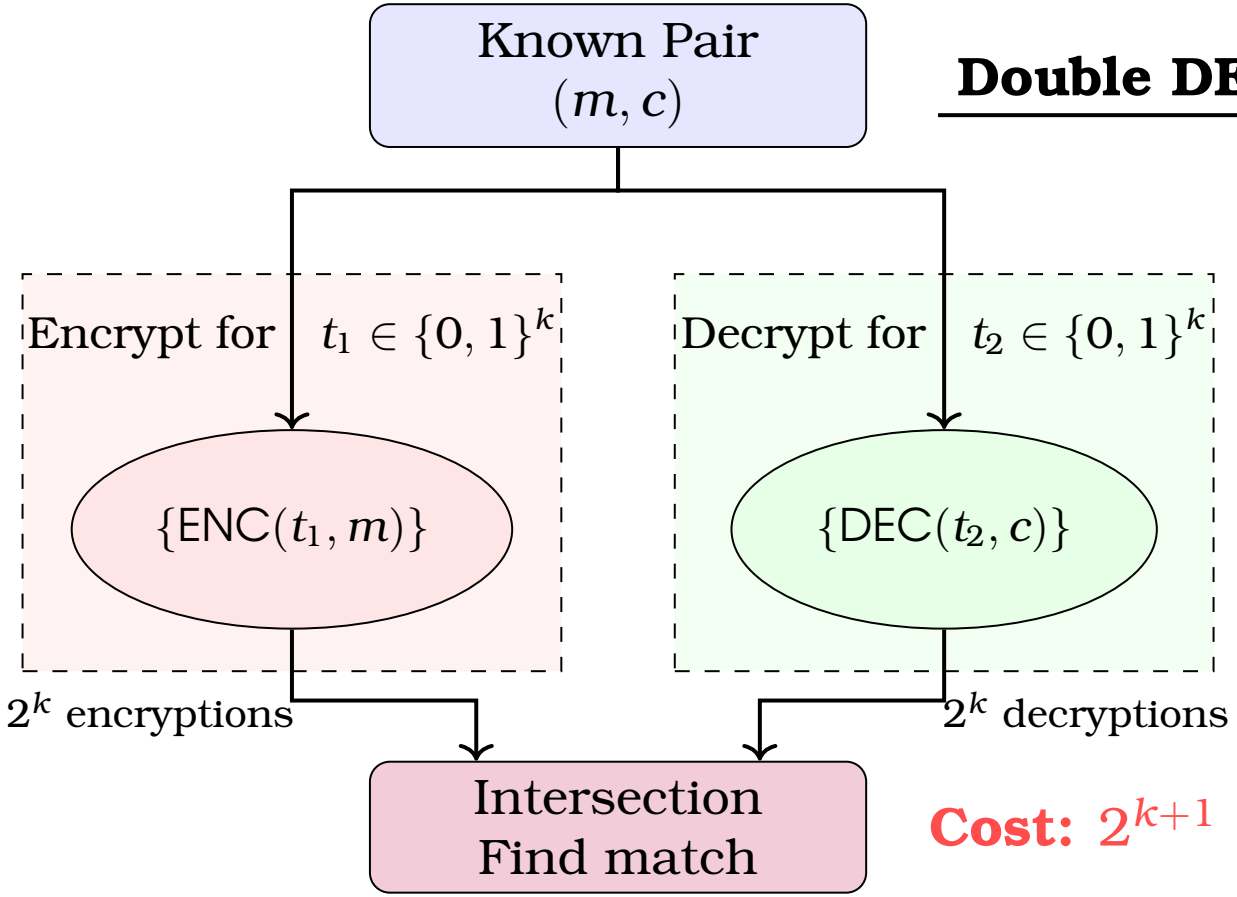

Double DES Meet-in-the-Middle

Cost: 2^{k+1} operations

What now?

If short scalars are not used, minimal efficiency is gained. If short scalars are used, no unconditional security can be proven...

Remark: Once the protocol execution is finished, $\mathcal{A}$ can no longer cheat. We could aim for everlasting security instead.

(Unorthodox) Solution:

- Add a timeout parameter to the protocol.
- Bound $\mathcal{A}$ computationally... but not in the conventional way: Assume $\mathcal{A}$ is not more powerful than the entire Bitcoin network, i.e., $\mathcal{A}$ is allowed to compute 2^κ scalar group multiplications per second (κ matching the current hashrate conversion).